

BRAC UNIVERSITY

Department of Computer Science and Engineering

Examination: Midterm
Duration: 80 Minutes
No. of Questions: 4

CSE 220: Data Structures

Semester: Spring 2026
Full Marks: 30
No. of Pages: 3

Name:	ID:	Section:
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- ✓ Use the back **part** of the answer script for rough work. **No washroom breaks.**
- ✓ At the end of the exam, put the question **paper** inside the answer script and **return both.**
- ✓ **Understanding questions is part of the exam.**

[You're not allowed to use any extra data structures to solve Question 1 & 2]

Question 1: CO5 [6 Points]

The hidden message was encrypted by placing characters in a 2D matrix, and it can only be recovered if the matrix is traversed in a specific order. Your task is to write a method **decode()** that takes the matrix in its parameter and follows the given traversal sequence to print the decoded message. Assume, the column length of the matrix is always even.

Sample Given Matrix	Sample Output																		
<table><tr><td>T</td><td>A</td><td>E</td><td>S</td><td>—</td><td>H</td></tr><tr><td>V</td><td>R</td><td>E</td><td>R</td><td>E</td><td>T</td></tr><tr><td>G</td><td>—</td><td>I</td><td>R</td><td>!</td><td>D</td></tr></table> Hint: Imagine a 2x2 block traversing the 2D matrix row-wise. To keep track of visited/printed index, after printing assign '#' in that particular index so that it is not printed the second time.	T	A	E	S	—	H	V	R	E	R	E	T	G	—	I	R	!	D	TRAVERSE_THE_GRID! Explanation The sequence in which the grid is to be traversed (0,0) (1,1) (0,1) (1,0) → TRAV (0,2) (1,3) (0,3) (1,2) → ERSE (0,4) (1,5) (0,5) (1,4) → _THE (1,0) (2,1) (1,1) (2,0) → _G (since (1,0)&(1,1) were printed earlier, it was not printed the second time, so any index/co-ordinate visiting the 2nd time will not be printed) (1,2) (2,3) (1,3) (2,2) → RI (1,4) (2,5) (1,5) (2,4) → D!
T	A	E	S	—	H														
V	R	E	R	E	T														
G	—	I	R	!	D														
Python Method Signature	Java Method Signature																		
<pre>def decode(matrix):</pre>	<pre>public static void decode(char[][] matrix)</pre>																		

Question 2: CO1 [8 Points]

Assume a custom Non-Dummy headed linear Doubly Linked List has been implemented using a class called **CellNode**. Each node contains an integer **val** and an integer **pulse_radius**. A **pulse_radius** greater than zero indicates that the node emits a pulse. Pulses occur one after another from left to right. When a node emits a pulse, it destroys itself, a number of nodes to its immediate left and right equal to its **pulse_radius**. Remaining nodes "snap" together, re-linking to maintain the list structure.

If a node containing a pulse is destroyed by a previous pulse, it is gone and cannot emit a pulse separately and does not affect its neighbors. After a pulse, the process continues with the next available pulse node to the right of the destroyed zone. If the head of the list is destroyed, the first available surviving node becomes the new head. Your goal is to complete the function **sendPulse(CellNode head)** that takes the head of the linked list and returns the head of the final linked list after all pulses have occurred.

Python Method Signature	Java Method Signature
def sendPulse(head):	public static CellNode sendPulse(CellNode head)

Python CellNode Class	Java CellNode Class
<pre>class CellNode: def __init__(self, val, radius): self.val = val self.pulse_radius = radius self.next = None self.prev = None</pre>	<pre>class CellNode { int val; int pulse_radius; CellNode next; CellNode prev; }</pre>
Sample Given LinkedList	Sample Returned LinkedList
(5,0) <-> (10,1) <-> (15,0) <-> (20,0)	(20, 0)
Explanation: The pulse at (10,1) bursts, destroying its left neighbor (5, 0), itself, and its right neighbor (15, 0). Since the first three nodes are removed, (20, 0) is the only survivor and becomes the new head.	
(5,0) <-> (10,0) <-> (15,1) <-> (20,1) <-> (25,0)	(5, 0) <-> (25, 0)
Explanation: The pulse at (15, 1) bursts, destroying its left neighbor (10, 0), itself, and its right neighbor (20, 1). The remaining survivors, (5, 0) and (25, 0), instantly snap together to form a new link.	

Question 3: CO1 [6 Points]

At an airport, passengers line up in a **queue** to board a flight. According to the boarding policy, passengers are processed in **groups of k**. Before each group boards, the airline staff **reverses the order of passengers within that group** for security verification. If the **last group contains fewer than k passengers**, their order remains **unchanged**. Write a function/method named **queueGroupReverse()** that reorganizes the queue according to this rule and **returns the updated queue**.

Assume that **linked-list-based Stack and Queue are already provided**. The **Stack** class provides the public methods **push()**, **pop()**, **peek()**, and **isEmpty()**, while the **Queue** class provides the public methods **enqueue()**, **dequeue()**, **peek()**, and **isEmpty()**. The instance variables of these classes are **private**.

You **cannot use any data structure other than the given Stack and Queue**. However, you may create **multiple stacks or queues if necessary**.

Java Method Signature		Python Function Signature
public static Queue queueGroupReverse(Queue q, int k)		def queueGroupReverse(q, k):
Sample Given Queue	Sample Returned Queue	Explanation
Queue: 1 2 3 4 5 6 7 8 k = 3 leftmost is the front	(front) 3 2 1 6 5 4 7 8	Passengers are processed in groups (1 2 3), (4 5 6), and (7 8). The first two groups are reversed, while the last group contains fewer than k passengers and therefore remains unchanged.
Queue: 5 8 11 14 17 k = 4 leftmost is the front	(front) 14 11 8 5 17	Passengers are processed in groups (5 8 11 14) and (17). The first group is reversed, and the last passenger remains in the same position since the group size is less than k.

Question 4: CO3 [10 Points]

Answer the questions in your Answer Script

I. A hash table of size 10 is used to store the following keys: 15, 25, 35, 45, 55, 65, 75.

Assume forward chaining will be used for collision handling. Following four hash functions are proposed:

$H1(k) = k \bmod 10$	$H2(k) = (k / 10) \bmod 10$	$H3(k) = (k + 5) \bmod 10$	$H4(k) = (k * 2) \bmod 10$
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All the results will be in integer, if there are fractions the floor value is considered.

Which hash function is the best choice here to reduce the collision?

[1 mark]

- (a) H1 (b) H2 (c) H3 (d) H4

II. What is the time complexity of traversing a linked list containing n nodes?

[1 mark]

- (a) $O(\log(n))$ (b) $O(n)$ (c) $O(n^2)$ (d) $O(n/2)$

III. What data structures can not be used to reverse a queue?

[1 mark]

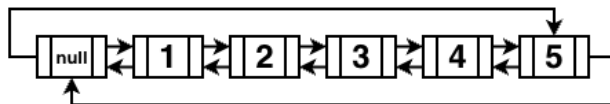
- (a) Array (b) LinkedList (c) Queue (d) Stack

IV. Given a stack, three push functions push(1), push(2), push(3) are being called. However, in between these push() functions or at the end of calling these push() functions, three pop() functions are being called. Which sequence of outputs can never be generated given this scenario?

[1 mark]

- (a) 3 2 1 (b) 3 1 2 (c) 2 1 3 (d) 1 2 3

V. Given the following dummy-headed doubly circular linked list, where the dummy head node is the node having null value. The **modify(dh)** function is executed with dh(dummy head) passed as the argument:



function modify(dh):

loop 2 times:

n1 = dh.next

n2 = n1.next

n3 = dh.prev

dh.next = n2

n2.prev = dh

n3.next = n1

n1.prev = n3

n1.next = dh

dh.prev = n1

After the function finishes executing, draw the resulting linked list in the answer script. [3 mark]

VI. Which linked list's head variable points to a node even when the list is empty?

[1 mark]

- (a) Doubly Linked List
(b) Circular Linked List
(c) Singly Linked List
(d) Dummy Headed Linked List

VII. If a linear array of 500 length has been converted into a multi dimensional array of 10x5x10 shape. What would be the multi dimensional index of linear index 341?

[2 mark]